

Personal Authentication Based on Iris Texture Analysis

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Abstract

We present an iris identification approach based on wavelet-packet binary coding. This latter is carried out by coding the 64 wavelet-packet energies to generate a compact signature (iris code), quantizing these energies into one bit according to an adapted threshold. Experimental results, using CASIA database, are presented demonstrating significant improvements in iris verification accuracy.

1. Introduction

For the last decade, biometric-based personal verification and identification methods have gained much interest with an increasing emphasis on security. Iris recognition is a fast, accurate and secure biometric technique that can operate in both verification and identification modes since the iris texture pattern has no links with the genetic structure of an individual and since it is generated by chaotic processes [1]. Consequently, a variety of iris recognition approaches were proposed that can be broadly classified in three categories depending on the method used to extract features from the texture. These categories are: (1) texture-based, (2) appearance-based and (3) feature-based extraction. Texture-based techniques [2-12] make use of filters for image processing and extract iris features from the filtered images. Daugman [2] used Gabor wavelets to extract texture phase structure information of the iris to compute a 256-byte binary iriscode. The Hamming distance measure was used to compare the difference between a pair of iris representations. In Wildes' approach [4,5], the Laplacian pyramid constructed with four different resolution levels is used as iris features. Then, the normalized correlation is used to compare both the input and the template iris images. Wang et al [8, 9]

have proposed recognition techniques using multichannel Gabor filtering. Hence, they have used 8-directional Gabor filters with multiple frequencies to capture both global and local details in an iris image. The mean and variance of these Gabor filtered images are used as features for the matching process. Other approaches that exploit other wavelet categories such as the 2D Haar wavelet and zero-crossing representation of 1D wavelet Transform, were proposed in [9,10] to characterize an iris texture.

Appearance-based approaches [11-15], such as Independent Component Analysis (ICA), direct Linear Discriminant Analysis (LDA) and Principal Component Analysis (PCA), attempt to use classical statistical approaches to extract iris features. The PCA is superior in image construction, because it can control construction errors by selecting the cumulative variance. Euclidean-distance and Nearest-Neighborhood (NN) classifier are adopted in these approaches.

Feature-based extraction techniques [8, 16,17] use local variations, which are characterized by the appearance and disappearance of an important image structure. A bank of spatial filters is constructed for efficient feature extraction and for matching process.

However, if the original iris images are of poor quality, the wavelet-packet approaches are more efficient for iriscode generation comparatively to the classical wavelet approaches without a greater risk of iris recognition errors. For this reason, some authors have proposed the iris recognition based on wavelet packet [19].

In this paper, we present a method for iris recognition based on a wavelet packet decomposition of iris images. Each iris image is described by a subset of band-filtered images (subimages) containing wavelet coefficients. From these coefficients, which characterize the iris texture, we compute a compact iris

feature code using the appropriate energies of these subimages to generate binary iris codes according to an adapted threshold. Thereafter, we show how an efficient and reliable Hamming distance can be used in order to classify iris codes. Results are presented that demonstrate significant improvements in iris recognition accuracy through the use of the public iris database CASIA which is available in [21]. These results are compared to those obtained in [18, 19].

This paper is organized as follows: the iris feature extraction is mentioned with details in Section 2. The matching process is described in Section 3. Experiments carried out are reported in Section 4.

2. Iris feature extraction

A typical iris feature extraction method has two major components: (a) iris image segmentation and (b) iris feature code generation.

2.1. Iris image Segmentation

The iris is an annular part between the pupil (inner boundary) and the sclera (outer boundary). Different iris localization approaches have been reported in the literature: Integro-Differential, Circular Hough, Hough Transform and edge and contour detection. In our approach, this task is performed by detecting the image intensity changes using the Integro-differential operator [2] which exploits two principal steps: (1) Define papillary boundary, limbic-iris boundary and define upper and lower eyelid boundary(Figure1b), (2) Unwrap the circular iris portion to a rectangular image block(Figure 1c). For iris feature vector extraction, we use the approach introduced in [22].

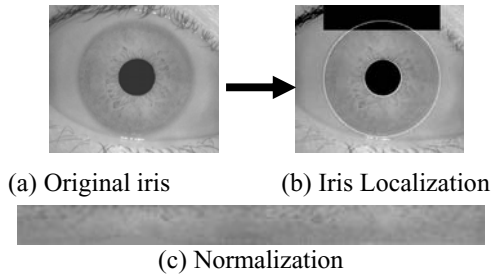


Figure 1. Iris image segmentation

2.2. Iris Feature Code Generation

The majority of researchers have proposed wavelets approaches in order to capture local iris features at different scales. In our case, we have adopted the method introduced in [19].

After the iris image segmentation process is completed, the iriscode is performed using Haar

wavelet packets as well as the energy of the packets sub-images to extract texture phase structure information of the iris and to compute the iris 64-bits codes.

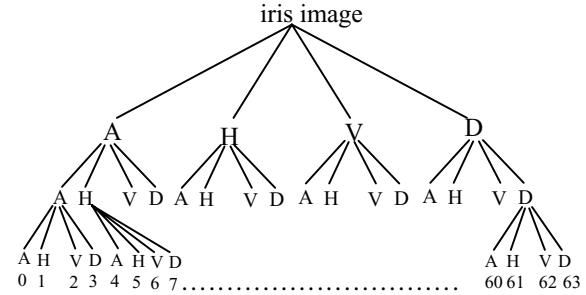


Figure2. Wavelet packets decomposition for iris image

The Iris code generation process can be summarized in the following steps:

1. *Wavelet packets decomposition*: We have used the Haar wavelet in a 3-level wavelet packet decomposition to extract the texture features of the unwrapped images [18]. This generates 64 wavelet packets (output iris subimages), numbered 0 to 63.

The images contain approximation (A), horizontal detail (H), vertical detail (V) and diagonal detail (D) coefficients respectively as shown in Figure 2.

2. *Wavelet packets energy computation*: In order to obtain the most texture information in packet subimages, we have used an energy measure. The mean energy distribution allows evaluating which packets are used to compute the normalized adapted threshold for our iriscode generation. The energy measure E_i for a wavelet packet subimage W_i can be computed as follows [19]:

$$E_i = \sum_{j,k} W_i(j,k)^2 \quad (1)$$

Figure 3(a) plots the mean energy distribution for the 64 wavelet packets subimages shown in Figure 1. This figure shows that energies of the wavelet packets 1 and 49 are the appropriate energies and the most suitable to compute the adapted threshold for iris code generation. However, in Figure 3(b), the energies of the wavelet packets 1 and 3 are the appropriate energies. On the other hand in [18], the authors have mentioned that the packet 2 and 10 are always used to compute their iriscode but, the most texture information are located in packets (subimages) 1 and 49. This shows that the author's assumption for the iris

signature computation using wavelet packets 2 and 10 is not always true because they have been using the mean energy distribution for the 149 unwrapped images of their iris database. This result has been confirmed by the authors of [21] who have mentioned that the *HD* distribution for the CASIA database has not been performed yet using the approach [19]. Moreover, Table 1 shows the first four maximal energy values for the 64 subimages for each iris specimen of the Image 088 of CASIA database. Clearly, we have not the same dominant energy for each specimen image iris. Figure 4 shows those packets 2, 49 and 7 holds much energy as well for the 756 CASIA iris images. It is therefore interesting to evaluate also the possibilities of using those packets as appropriate energies to generate iris codes as described in [19].

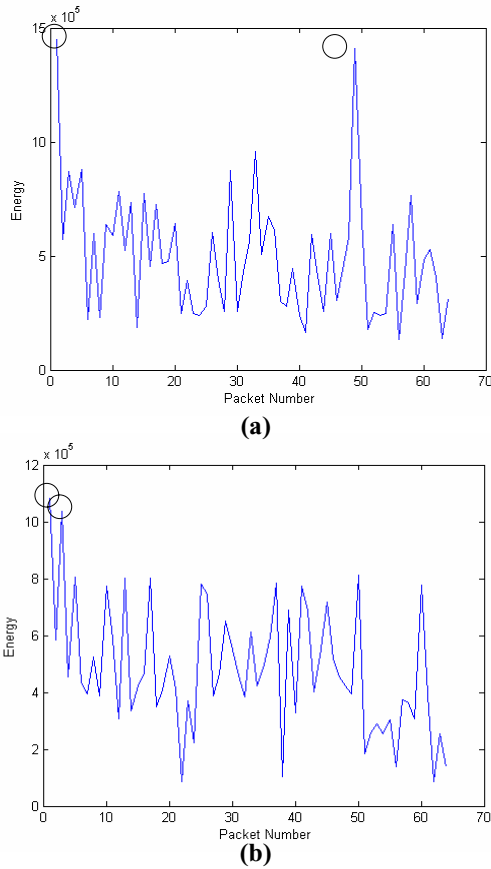


Figure 3. Energies wavelet packet distribution for (a) Iris 088_1_1 and (b) Iris 088_2_5.

Table 1. The appropriate energy values for iris CASIA subimages 088.

CASIA iris specimen images 088							
Packet number	1	2	3	4	5	6	7
	49	49	2	7	3	35	7
	35	3	35	35	50	3	29
	2	11	49	49	46	49	3

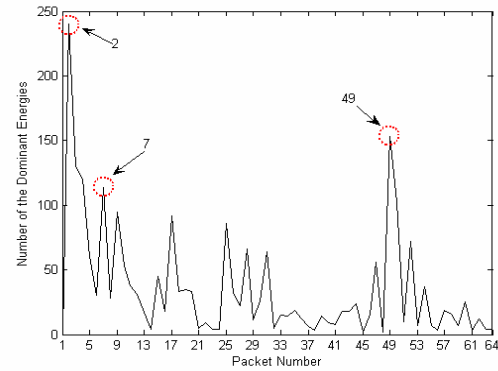


Figure 4. Mean wavelet packet energy distribution for 64 packets.

In our approach, we do not adopt the same strategy as that in [19]. We use the appropriate wavelet packet energies of each iris image to compute the adapted threshold to encode the 64 subimages.

Let $E_1 \dots E_\lambda$ be the appropriate wavelet packet energies of the packets $1 \dots \lambda$ respectively. We define the normalized adapted threshold S as follows:

$$S = Coeff \cdot \frac{\mu(E_1, \dots, E_\lambda)}{\text{Max}(E_1, \dots, E_\lambda)} \quad (2)$$

where $\mu(E_1, \dots, E_\lambda)$ represents the mean wavelet peak energy value, $Coeff$ as a constant and λ is the number of the appropriate energies.

Therefore, the threshold S is not optimized experimentally ($S=10$) as defined in [19] for all iris images but each iris wavelet decomposition produces the adapted threshold S .

For example, from Table 2, we have several adapted thresholds to encode the iris images. So, for iris 088_1_1, we choose $S=1.341$ to code the subimages using only the two maximal energies of the packets 1 and 49. But we choose $S=1.320$ using the three first maximal energies for the packets 1, 49 and 35. However, for another iris image we will not have the same adapted thresholds. In this paper, we have choose $Coeff=1.4$ as a constant.

Table 2. The adapted thresholds for iris CASIA images 088

	Iris images	Number of appropriate energies	
		$\lambda=2$	$\lambda=3$
Adapted threshold	1	1.341	1.320
	2	1.170	1.086
	3	1.395	1.350
	4	1.397	1.396
	5	1.393	1.389
	6	1.390	1.335
	7	1.294	1.247

3. *Iris feature coding*: After determination the appropriate wavelet packets energies and the normalized adapted threshold, we can carry out the coding of the 64 wavelet packets energies to generate a compact iriscode by quantizing these energies into one bit according to each appropriate energy. Let E_λ be the appropriate energy of the peak λ . Then the iriscode C_λ computed according to E_λ we define by the following:

$$C_\lambda(j) = \begin{cases} 1 & \text{if } \frac{E_j}{E_\lambda} > S \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Where $j = 0 \dots 63$.

In our approach, we have used significant wavelet coefficients of the iris subimage. Each used appropriate energy resulting in a total of 64 bits which correspond to the 64 subimages of the iris wavelet decomposition. Therefore, we obtain one iriscode according to each energy. The iriscode of the iris 088_1_1 of Fig 2, computed using the three first energy peaks ($\lambda = 2 \dots 4$) is illustrated in Figure 5. We note that our iris coding gives distinct iriscodes for each appropriate energy E_λ .



Figure 5. Iriscodes according to packet energies.

3. Iriscodes matching

The iris codes matching task is performed by pairing the iriscodes extracted from the input and the template iris images. The most common comparison method of iris signatures is the **Hamming Distance**

(HD) [12]. This comparison is to be made with the user's iris template, which will be calculated depending on the binary matching algorithm which is implemented by the Boolean exclusive-OR operator ($\oplus$) applied to the 64 bit vectors that encode any two iriscodes.

Let $T = \{C_i(j) \quad j=0 \dots N\}$ and $I = \{C_k(j) \quad j=0 \dots N\}$ be the iris binary codes extracted from the template and input iris images. The binary Hamming distance $HD_{i,k}$ between these iris binary codes is measured according to the following [12]:

$$HD_{i,k} = \frac{1}{N} \sum_{j=0}^{j=N} C_i(j) \oplus C_k(j) \quad (4)$$

The preceding distance is computed according to the appropriate energy (maximal value) between two iris signatures in order to decide if both come from images of the same iris (authentic distance) or from images of different iris (impostor distance). Therefore, the combination of two or several packets is likely to hold more discriminating information than what each of them does separately as mentioned in. So, the preceding distance can be extended for two appropriate energies of each iris image as follow [18]:

$$HD_{i,k} = \frac{1}{2 * N} \sum_{j=0}^{j=N} \|C_i(j) \oplus C_k(j)\| \cdot \|C_i(j) \oplus C_k(j)\| \quad (5)$$

where C_i and C_k are the iris codes computed according to the two appropriate energies of the template iris image and C_k and C_k of the input one.

4. Experimental results

In this section, we analyze the performance of the proposed iris verification algorithm. So, to evaluate the performance of our algorithm, different experiments on the Institute of Automation, Chinese Academy of Science, CASIA [20] database have been carried out. As mentioned in [23,24], automatic detection of the pupils is easier because CASIA database images have a black circle in the pupil and some of these black circles cover the part of iris and also the specular reflection. First, we report the experimental results obtained running our algorithm using several wavelets categories. Thereafter, we present some result to study the effect of the wavelet packet combination. Finally, we compare our performance with the algorithms reported in [18,19] which use the fixed threshold.

For the first experiment, we have uses the first wavelet packet feature that has the maximal energy

value to generate the iris signature (to encode 64 bits of the wavelet packets). This experiment was motivated by an iris code computation using only two appropriate energies according to the equation (2 and 3) to compute the adapted threshold and one iriscode according to the maximal energy. The energy corresponding to subimage 0 has not been used because this subimage contains offset information.

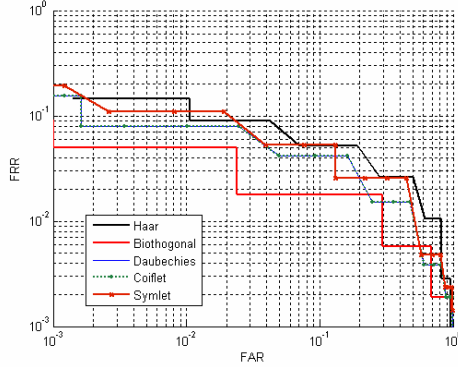


Figure 6. The ROC of iris recognition on CASIA database according to various wavelets.

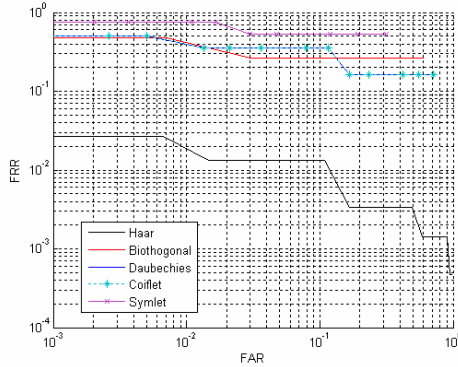


Figure 7. The ROC of iris recognition on CASIA database according to appropriate energies combination.

The discrimination results using wavelet packets categories are shown in Table 2. Clearly, using the first appropriate energy, the Haar wavelets are most suitable for iris recognition. The Receiver Operating Curve (ROC) for results is shown in Figure 6 (FAR: False Acceptance Rate and FRR: False Reject Rate).

Thus, our matching algorithm is slower with the error rate EER=2,38%, 4.67%, 4.68%, 5.28% and 6.28% for Biothogonal, Daubechies, Symlet, Coiflet and Haar wavelets, respectively.

However, the appropriate energies of the packets combination have independently proved to achieve positive separation only for the Haar wavelets compared to the choices of these appropriate energies.

Figure 7 gives the performance results using the dominant energies combination. Obviously, we can see consistent performance improvement of our matching using the first and the third energies of each iris image decomposition. Hence, using the combination of these appropriate energies for iriscode generation, our matching algorithm is slower with the error rate EER=1.41% using Haar wavelets. The drawn significant result starting from these experiments is that the combination of the wavelet packets for the other wavelets types does not give a better performance for our iris recognition system.

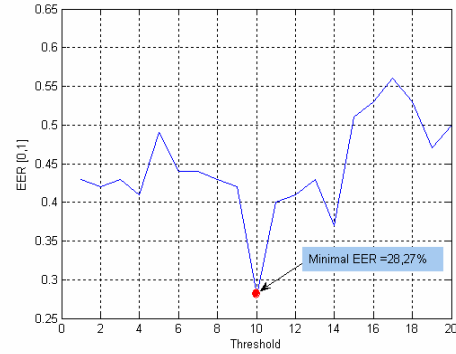


Figure 8. The threshold optimized experimentally in the approach proposed in [19].

Before comparing our approach with those of [18,19], we illustrate in Figure 8, the minimal ERR curve by varying the threshold S from 1 to 20. With the threshold $S=10$, we obtain the minimal EER for the CASIA database. Clearly, the matching algorithm [19] achieves better performance for the threshold $S=10$ using the packets 2 and 49 with EER=28.27%. This confirms the choice of the authors.

Table 3. The comparative results on CASIA database.

	Number of Packets	EER (%)	Iriscode size
Our approach	1	6.28	64
	2	1.41	128
	3	0.33	196
Approach [19]	1	32.45	512
	2	28.27	1024
	3	28.89	1536

Compared with these approaches, our iris matching achieves better performance using one appropriate energy or the combination of several ones. The comparative results are shown in Table 3 in terms of iris code size and ERR in respect to the database

CASIA. Obviously, the experiments conducted over a sample of all CASIA iris image database indicate that the improving matching accuracy is not directly proportional to the combination of the iris codes generated from the appropriate energies as in the approach [19]. As illustrated in Figure 9, we have obtained the error rate EER=1.41%, 3.71% and 4.76% using a single appropriate energy and EER= 3.03%, 1.41% and 5.0% for the combination of the first three appropriate energies, respectively. Figure 9 shows the ROC curves comparison of our approach with the ones proposed in [19] in both these situations. It can be seen that, in all cases, the Equal Error Rate, is always below 5%, achieving 1.05% for the first energy. More importantly a FAR=0.06% has been obtained for very low rates of False Rejection FRR=1.2%, which means that our system is more appropriate for very high security environments.

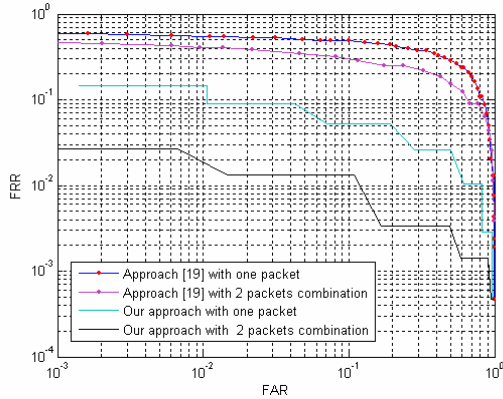


Figure 9. The ROC comparison of iris recognition on CASIA database.

5. Conclusion

The primary focus of this work is a personal authentication system based on human iris verification using wavelet packets decomposition. The proposed technique uses only appropriate packets with dominant energies to encode iris texture according the adapted thresholds. The usefulness of this approach was confirmed in the experiments conducted here, which reveals that the verification results with an EER=0.3% has been obtained for packets combination, which means that our system is appropriate for very high security environments.

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7. References

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